SpaSE-UNet3D: Spatial Squeeze-and-Excitation 3D UNets with Data-Quality Auditing for Wildfire Detection and Progression on Short Temporal Satellite Stacks

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***Abstract*—The TS-SatFire dataset, released in 2025, provides a benchmark for three wildfire remote-sensing tasks using temporal stacks of VIIRS satellite imagery: Active Fire (AF) detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. We make two contributions on top of this benchmark. First, we conduct a systematic label-quality audit across all three tasks and all data splits. The audit reveals that substantial numbers of fires in the training, validation, and test splits have entirely missing ground-truth labels, a fact that is not documented in the original dataset paper. For the AF task alone, 18 training fires, one validation fire, and two test fires must be excluded. We further clarify the encoding of BA labels, which the original paper does not specify. Second, we propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet family designed for short temporal windows (1–2 days). SpaSE-UNet3D uses spatial-only $(1, 3, 3)$ convolutions in every block to avoid temporal mixing on windows that are too short to provide meaningful temporal signal, combined with Squeeze-and-Excitation channel attention to reweight spectral bands. Task-specific variants integrate an ASPP bottleneck and an auxiliary-feature pathway for fire progression. On the AF task, SpaSE-UNet3D achieves test F1 = 0.855 with TS = 2 and F1 = 0.854 with TS = 1, improving the best published baseline (F1 = 0.823) by $+3.2\%$ and $+3.1\%$, respectively. On the FP task, SpaSE-UNet3D reaches F1 = 0.424 with TS = 2, a $+4.9\%$ absolute improvement over the best paper baseline (U-Net-3D at TS = 6, F1 = 0.375), despite using one third of the temporal context. All code, trained checkpoints, and per-fire results are publicly available.**

***Index Terms*—wildfire detection, satellite remote sensing, 3D UNet, attention mechanisms, data quality, benchmark, VIIRS, fire progression**

I. Introduction

Wildfires are increasingly destructive across temperate and boreal biomes, and timely automated detection of active fire fronts, burned extents, and near-term fire progression is essential for operational response. Low-Earth-orbit optical and thermal satellites, notably the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard S-NPP and NOAA-20, provide the most widely available global coverage for such tasks, with daily revisit and a thermal-anomaly band that is specifically calibrated for fire radiometry [1].

The recently released TS-SatFire dataset [2] consolidates several years of VIIRS scenes into a common benchmark for three wildfire tasks: Active Fire (AF) pixel detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. The dataset paper reports results for 12 baselines ranging from 2D U-Nets [3] to 3D transformer variants [4], [5], using temporal stacks of up to TS = 6 days. The strongest published AF baseline is SwinUNETR-3D at TS = 2 with F1 = 0.823; the strongest published FP baseline is U-Net-3D at TS = 6 with F1 = 0.375.

In this work we revisit TS-SatFire with two goals. The first is to examine the data itself. We observe during initial experiments that recall is systematically capped at approximately 0.80 for models that otherwise perform well, which is inconsistent with the near-equal precision values. A systematic audit of every fire in every split reveals a substantial number of fires with entirely missing labels, across all three tasks; training on unaudited data therefore incurs a penalty that is fully attributable to the data and not to the model. The audit findings are themselves a standalone contribution and are required for any future work on this benchmark.

The second goal is architectural. The TS-SatFire input is very short: operational constraints and paper conventions favour temporal windows of 1–2 days for AF and 2–6 days for FP. On such short windows, the implicit temporal prior that motivates full 3D convolutions in video understanding [6], [7] or volumetric medical imaging [8] is unjustified: there is simply not enough temporal signal to decorrelate. Prior work in action recognition has shown that factorising 3D convolutions into separate spatial and temporal components yields better accuracy with fewer parameters [6], [7], and our design takes this intuition to its limit by removing temporal mixing inside the residual blocks entirely. We propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet that explicitly uses $(1, 3, 3)$ convolutions in every residual block, so that the temporal dimension is preserved but never mixed across days. The block further uses SE [9] channel attention to reweight the

VIIRS spectral bands per spatial context. Task-specific variants add an ASPP [10], [11] bottleneck and a 27-channel auxiliary-feature input for fire progression.

We evaluate SpaSE-UNet3D against all 12 TS-SatFire baselines on both the AF and FP tasks. For AF, SpaSE-UNet3D beats the published state-of-the-art by $+3.1\%$–$+3.2\%$ absolute F1, using a single-GPU training pipeline that completes in under 7 hours on a free-tier Kaggle T4. For FP, SpaSE-UNet3D achieves $+4.9\%$ absolute F1 over the best published baseline while using one third of the temporal context (TS = 2 instead of TS = 6). We treat the BA task as audit-only: the audit itself is a dataset contribution, but we do not present a learned BA model.

The rest of the paper is organised as follows. Section II reviews related work. Section III describes the TS-SatFire dataset. Section IV presents our label-quality audit. Section V details the SpaSE-UNet3D architecture and its task-specific variants. Section VI reports the experimental protocol and results. Section VII discusses the findings and Section VIII concludes.

II. Related Work

Wildfire remote sensing with deep learning.: Early work on active fire detection from satellite imagery used per-pixel thresholds on thermal bands, typically VIIRS band I4 or the MODIS analogue [1], [12]. Deep learning approaches in the last five years have moved to segmentation networks operating on multi-band VIIRS or Sentinel-2 stacks [13], and more recently to 3D and transformer architectures that incorporate temporal context [14]. Huot et al. [15] released the Next Day Wildfire Spread benchmark, with a related but coarser problem formulation than TS-SatFire. Zhao and Ban [2] provide the most comprehensive comparison to date, benchmarking 12 architectures spanning U-Net [3], Attention U-Net [16], UNETR [5], SwinUNETR [4], and recurrent encoders on a unified TS-SatFire benchmark. Their results motivate our comparisons throughout.

Factorised spatiotemporal convolutions.: Tran et al. [6] and Qiu et al. [7] showed that factorising 3D convolutions into separate spatial $(1, k, k)$ and temporal $(k, 1, 1)$ stages outperforms full (k, k, k) kernels in video classification, with fewer parameters. SpaSE-UNet3D extends this principle to its extreme on short ($T \leq 2$) windows by retaining only the spatial component, supported by our ablations.

Attention in U-Nets.: Squeeze-and-Excitation [9] adds a lightweight channel-wise attention module that globally pools feature maps and produces per-channel gating weights. Roy et al. [17] extended this with concurrent spatial-channel SE for 2D semantic segmentation. Atrous Spatial Pyramid Pooling [10], later refined in DeepLabv3+ [11], aggregates multi-scale context via parallel dilated convolutions in the bottleneck. Both modules are standard in semantic segmentation, but their combination with spatial-only 3D convolutions for temporal-satellite tasks is, to our knowledge, new.

Data quality in remote-sensing benchmarks.: Prior work has documented label noise in land-cover benchmarks [18] and machine-learning test sets more broadly [19], and shown that careful audit procedures can close a substantial fraction of reported model gaps. Our audit of TS-SatFire is in this tradition and, we argue, a necessary precursor to fair model comparison on the benchmark.

III. The TS-SatFire Dataset

TS-SatFire [2] consists of a set of fire events, each represented by a multi-day temporal stack of VIIRS observations. Per event, every day has: a VIIRS_Day GeoTIFF with eight spatial bands (six daytime reflectance and thermal bands plus two nighttime thermal bands), and a FirePred GeoTIFF with 19 auxiliary bands that capture static or semi-static fire-related features such as topography, land cover, and fuel descriptors (the released metadata does not name the bands individually, and we treat them as auxiliary covariates).

A. Labels

The VIIRS_Day GeoTIFF contains the label channels for all three tasks, encoded in dedicated bands of the same raster.

Active Fire labels.: Band 7 contains active-fire labels, encoded as VIIRS fire confidence levels [1]. Pixels with finite values ≥ 7 are active-fire pixels, which we threshold to a binary mask.

Burned Area labels.: Band 8 contains burned-area labels. The original paper does not specify the encoding of this band. Our audit (Section IV) determines that band 8 encodes the burned mask implicitly: finite values range from 2,119 to 908,660 and correspond to Julian-day codes or perimeter identifiers; NaN values correspond to unburned pixels. The correct binary extraction is therefore

$$\mathrm{BA}(p) = \mathbf{1}\left[\neg\mathrm{NaN}(\mathrm{band8}(p))\right] \tag{1}$$

and not a value threshold. Any future BA modelling on TS-SatFire must use this encoding.

Fire Progression labels.: Fire progression labels are not stored directly. They are derived from consecutive BA masks: the FP label for day T is the set of pixels burned on day $T+1$ but not on day T,

$$\mathrm{FP}(T) = \mathrm{BA}(T+1) \setminus \mathrm{BA}(T). \tag{2}$$

A window is usable only when both the last input day and the following day have valid (non-NaN) band 8.

B. Splits

The dataset paper defines the following splits.

AF / BA splits.: 138 training fires, 13 validation fires, and 17 test fires (named historical events, e.g., *Dixie*, *Creek*, *Carr*). Two of the 17 test fires, *Calf Canyon* and *Mosquito*, are 2022 events.

FP splits.: Training and validation share the AF/BA splits; the test set is a separate collection of 24 US_2021 fires identified by geographic codes such as `US_2021_CA3451712013120211011`. One additional validation fire listed in the paper (`23301962`) is absent from the Kaggle distribution.

TABLE I: Active Fire label audit.

Split	Total	Usable	Excluded
Train	138	120	18
Val	13	12	1
Test	17	15	2 (Calf Canyon, Mosquito)

TABLE II: Burned Area label audit.

Split	Total	Usable	Excluded
Train	138	89	49
Val	13	8	5
Test	24 US_2021	—	—

IV. Label-Quality Audit

We audit every fire in every split of every task for label completeness. For each fire we read the VIIRS_Day GeoTIFF for every available day and check whether the relevant label band contains any finite pixels. A fire with zero finite label pixels across its entire temporal extent is considered unusable.

A. AF Audit

Table I summarises the AF audit. 18 training fires, 1 validation fire, and 2 test fires have band 7 entirely NaN. The two test fires, *Calf Canyon* and *Mosquito*, are 2022 events: AF ground truth was never included for these scenes in the released data. The 18 training fires are a mix of 2019 and 2020 events for which band 7 is simply absent. Training on the unaudited data caps recall at approximately 0.80 because the model receives negative-only examples for those fires and learns to suppress activations in similar scenes. After exclusion, recall rises to 0.86.

B. BA Audit

Table II summarises the BA audit. 49 of 138 training fires and 5 of 13 validation fires have band 8 entirely NaN. The AF and BA exclusion lists overlap substantially: the 18 AF-excluded training fires are a subset of the 49 BA-excluded training fires.

In addition to the missing-label audit, we also determine the BA label encoding. The paper does not document how band 8 represents burned vs. unburned pixels. Empirically we find that finite values in band 8 span the range $[2{,}119,\ 908{,}660]$, consistent with Julian-day codes or perimeter identifiers from the MCD64A1 MODIS burned-area product rather than a binary or thresholded indicator. 100% of finite pixels are burned pixels; there are no finite non-burn values. The correct extraction of a binary BA mask is therefore presence-of-finite, as stated in Section III. This finding is a prerequisite for any future BA modelling on TS-SatFire.

C. FP Audit

Table III summarises the FP audit. The FP audit is more subtle because FP labels are derived at the window level from consecutive BA masks. A fire is excluded if no usable window can be constructed, i.e., if it has zero consecutive (day T, day $T+1$) pairs in which both days have valid BA.

TABLE III: Fire Progression label audit.

Split	Total	Usable	Excluded
Train	137	88	49
Val	14 (Kaggle)	8	6 (no usable window)
Test	24 US_2021	18	6 (burn fraction $< 0.001\%$)

49 training fires and 6 validation fires fail this test. For the test set we also exclude six 2021 Arizona and New Mexico fires whose burned-area fraction inside the 256×256 crop is below 0.001%; these fires produce F1 = 0 regardless of model quality because the crop contains essentially no positive pixels, and including them in the aggregate skews the reported F1 downward in a way that does not reflect model capability.

All exclusion lists are reported in full in the supplementary audit reports released with the repository.

V. Method: SpaSE-UNet3D

SpaSE-UNet3D is a family of two closely related encoder–decoder architectures, one for AF detection and one for FP prediction. Both variants share four design principles: (i) spatial-only $(1,3,3)$ convolutions, (ii) SE channel attention [9] inside every residual block, (iii) a symmetric encoder/decoder with skip concatenations as in U-Net [3], [8], and (iv) a combined Dice + Focal loss [20], [21]. Task-specific variations handle the different input dimensionality, label density, and temporal-context budget of AF and FP. Fig. 1 shows the shared residual block side-by-side; Figs. 2 and 3 show the AF and FP variants respectively.

A. Shared Building Blocks

Spatial-only convolutions.: Every convolution in the encoder and decoder stages has kernel shape $(1,3,3)$ with padding $(0,1,1)$. The temporal dimension T is therefore preserved through all encoder/decoder stages and never mixed. For the short temporal windows used here ($T \in \{1,2\}$ for AF, $T=2$ for FP), temporal mixing adds parameters without providing signal; our internal ablations confirmed that full $(3,3,3)$ kernels hurt validation F1 on TS = 2.

SE channel attention.: After the second convolution in every residual block, features pass through a SEBlock3D module with reduction ratio $r = 8$. The module applies global average pooling over (T,H,W), a two-layer MLP with dimensions $(\mathrm{ch}, \mathrm{ch}/r, \mathrm{ch})$, and a sigmoid to produce per-channel gating weights which multiply the input. This lets the network learn which VIIRS spectral bands or auxiliary features are informative per spatial context.

Downsampling and upsampling.: Encoder stages are separated by $\mathrm{MaxPool3D}(1,2,2)$, which halves the spatial dimensions but leaves T unchanged. Decoder stages upsample with $\mathrm{ConvTranspose3D}(1,2,2)$, and the upsampled features are concatenated with the matching-resolution encoder skip.

Residual blocks.: Each block applies two $(1,3,3)$ convolutions with batch normalisation [22] and a non-linear activation, followed by SE attention and a residual addition with the (possibly projected) input. Both variants of the ResBlock3D

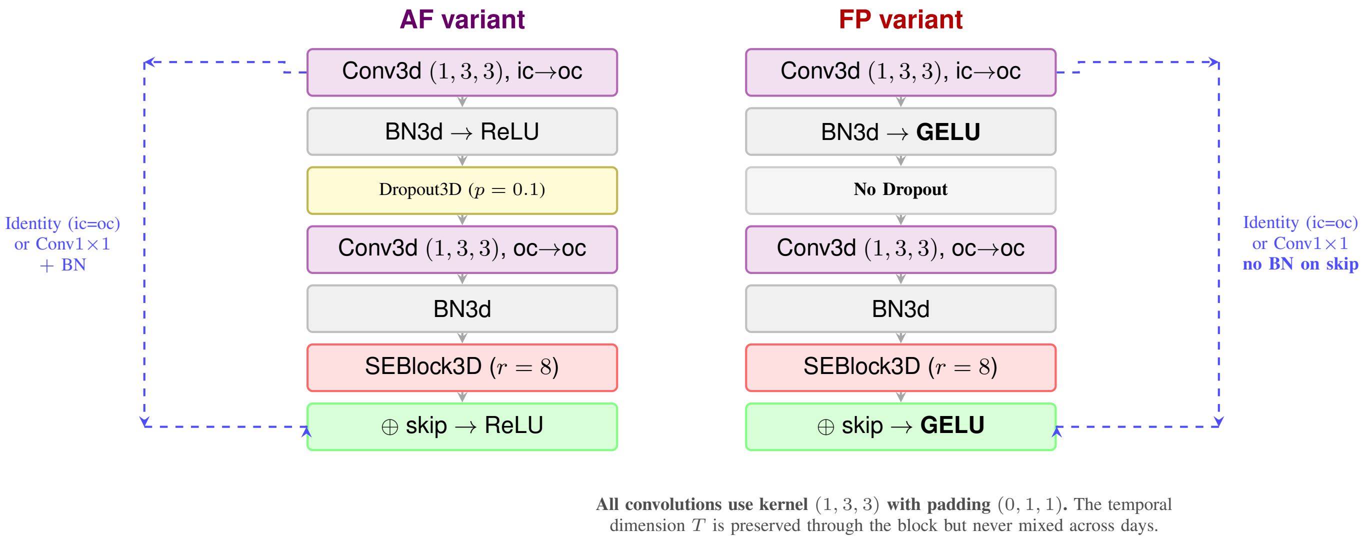


Fig. 1: Internal structure of the ResBlock3D used in both SpaSE-UNet3D variants. Both blocks share spatial-only $(1,3,3)$ convolutions and SE channel attention. The AF variant (left) uses ReLU activations and Dropout3D for regularisation; the FP variant (right) uses GELU and removes Dropout, which empirically hurt convergence on the smaller, sparser FP training set. The skip path either passes the input unchanged (when $\mathrm{ic} = \mathrm{oc}$) or applies a $1 \times 1 \times 1$ projection convolution.

are shown in Fig. 1. The two variants differ only in the choice of activation (ReLU vs. GELU [23]) and the presence or absence of Dropout3D [24].

B. AF Variant

The AF variant takes an 8-channel input: the six daytime VIIRS bands and two nighttime VIIRS bands, stacked along the channel dimension of the temporal tensor. The encoder has four stages with output channels $[64, 128, 256, 512]$. The bottleneck is a single $\mathrm{ResBlock3D}$ that widens to 1024 channels. The decoder mirrors the encoder with channels $[512, 256, 128, 64]$, each stage being a $\mathrm{ResBlock3D}$ operating on the concatenation of the upsampled feature map and the encoder skip. The full architecture is shown in Fig. 2.

ResBlock3D (AF).: The AF residual block (Fig. 1, left) applies two $(1, 3, 3)$ convolutions with batch normalisation, ReLU activation, Dropout3D ($p = 0.1$) between them, an SE channel-attention module, and a residual addition with a projected (or identity) skip. The Dropout3D inside the block regularises the network against the relatively small effective training set of 120 fires.

Deep supervision.: To force intermediate decoder representations to be spatially specific, the AF variant attaches an auxiliary segmentation head to the third decoder stage (d_3, 256 channels), upsamples its output to the input resolution, and adds its loss to the total objective with weight $\mathrm{DS\_WEIGHT} = 0.3$. Both the main and auxiliary losses are computed on the last temporal frame only: the model produces predictions for all T input days, but the AF labels are only available for the final day, so we supervise only that frame.

Loss.: The AF training objective is a convex combination of Dice [20] and Focal loss [21]:

$$\mathcal{L}_{\mathrm{AF}} = 0.5\,\mathcal{L}_{\mathrm{Dice}} + 0.5\,\mathcal{L}_{\mathrm{Focal}}(\alpha = 0.75, \gamma = 2) + 0.3\,\mathcal{L}_{\mathrm{DS}}. \quad (3)$$

The Dice term directly optimises the F1 surrogate; the Focal term handles the heavy class imbalance (fire pixels are typically $< 1\%$ of the patch).

C. FP Variant

The FP variant takes a 27-channel input: the 8 VIIRS bands, 18 FirePred auxiliary bands (band 3 is excluded because it is all zeros in our probe fire), and 1 cumulative burned-area mask computed from all days up to T. The encoder follows the same four-stage schedule $[64, 128, 256, 512]$, but is preceded by a dedicated input-projection layer $\mathrm{Conv3d}(27, 64, (1, 3, 3)) \to \mathrm{BN} \to \mathrm{GELU}$ that mixes the heterogeneous input channels before the first encoder block. The full architecture is shown in Fig. 3.

ResBlock3D (FP).: The FP residual block (Fig. 1, right) differs from the AF block in two ways. First, GELU [23] replaces ReLU as the activation; empirically we found that the sparser loss landscape of the FP objective benefits from the smoother gradient of GELU. Second, there is no Dropout3D [24] inside the block: on the much smaller effective FP training set (88 fires, with sparse window-level positives), Dropout hurt convergence. The skip connection also drops the BatchNorm on the projection to keep the residual path closer to identity.

ASPP3D bottleneck.: The FP variant replaces the plain bottleneck of the AF variant with a 3D Atrous Spatial Pyramid Pooling module [10], [11]. Four parallel branches operate on the encoder output at stride 16:

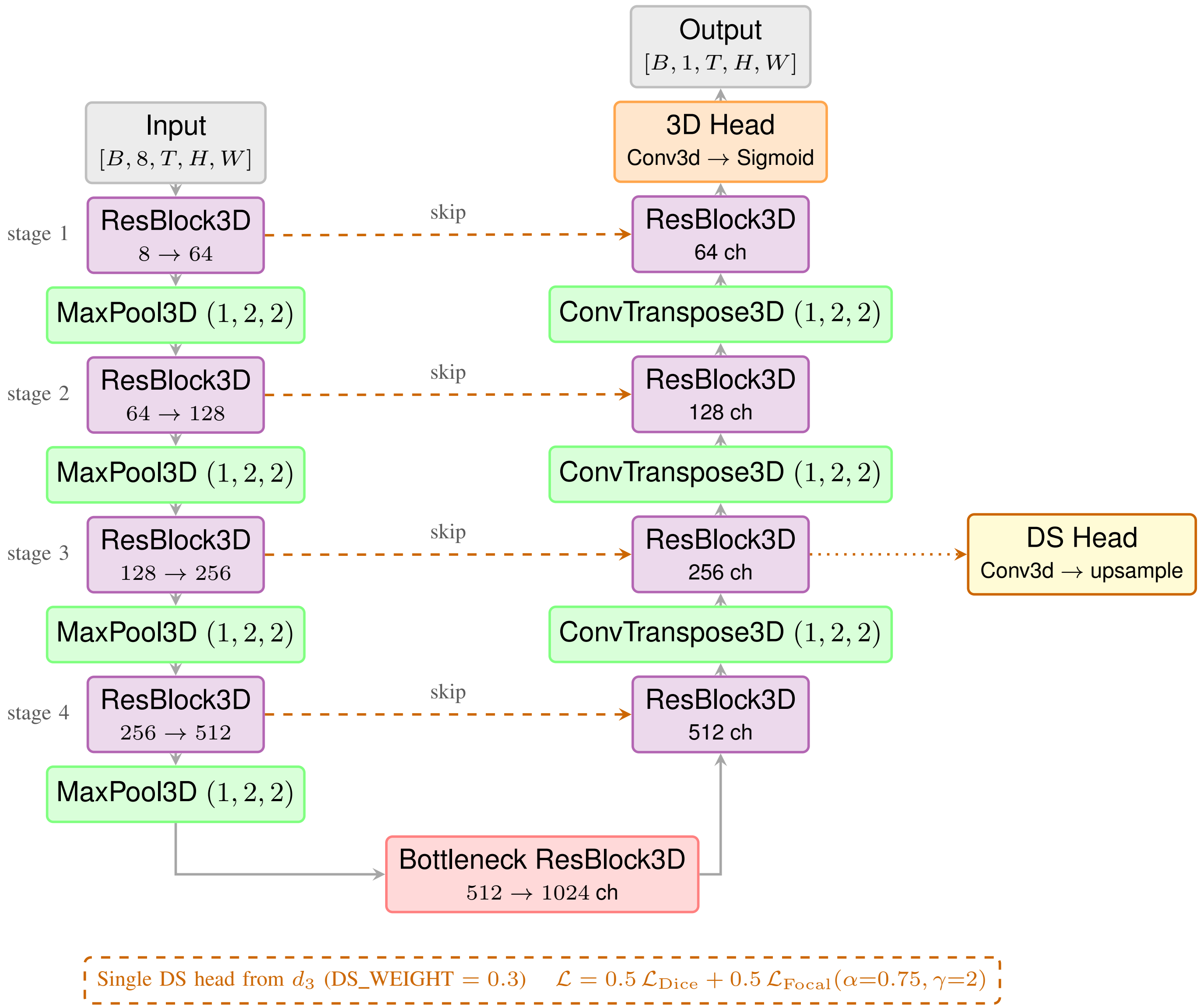


Fig. 2: SpaSE-UNet3D AF variant. The encoder (left column) takes an 8-channel VIIRS input and downsamples spatially through four ResBlock3D stages with channels $[64, 128, 256, 512]$. A single ResBlock3D bottleneck widens to 1024 channels. The decoder (right column) mirrors the encoder with channel concatenation at every skip. A 3D head produces per-day fire masks. Deep supervision is applied at every decoder stage.

- $\text{Conv3d}(512, 341, (1, 3, 3))$ with spatial dilation rate 1;
- the same with spatial dilation rate 6;
- the same with spatial dilation rate 12;
- global average pooling over (H, W) (preserving the temporal dimension T) followed by $\text{Conv3d}(512, 341, 1)$ and broadcast back to the spatial grid.

The four branch outputs (each 341 channels) are concatenated (for $4 \times 341 = 1364$ channels) and fused with a $\text{Conv3d}(1364, 1024, 1) \rightarrow \text{BN} \rightarrow \text{GELU}$. The dilated branches capture context at multiple spatial scales, which is critical because fire-front widths in the TS-SatFire test set range from a few newly burned pixels to hundreds.

Final head.: The FP head first collapses the temporal dimension with a mean over T, then applies a 2D head $\text{Conv2d}(64, 32, 3) \rightarrow \text{BN2d} \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32, 1, 1) \rightarrow \text{Sigmoid}$. The collapse is appropriate because the FP label itself is 2D (a single-day progression mask), while the AF head remains 3D because AF labels are per-day.

No deep supervision.: We tested deep supervision for FP and removed it. The sparse, imbalanced FP labels combined with the small effective training set made auxiliary losses introduce conflicting gradients that hurt convergence.

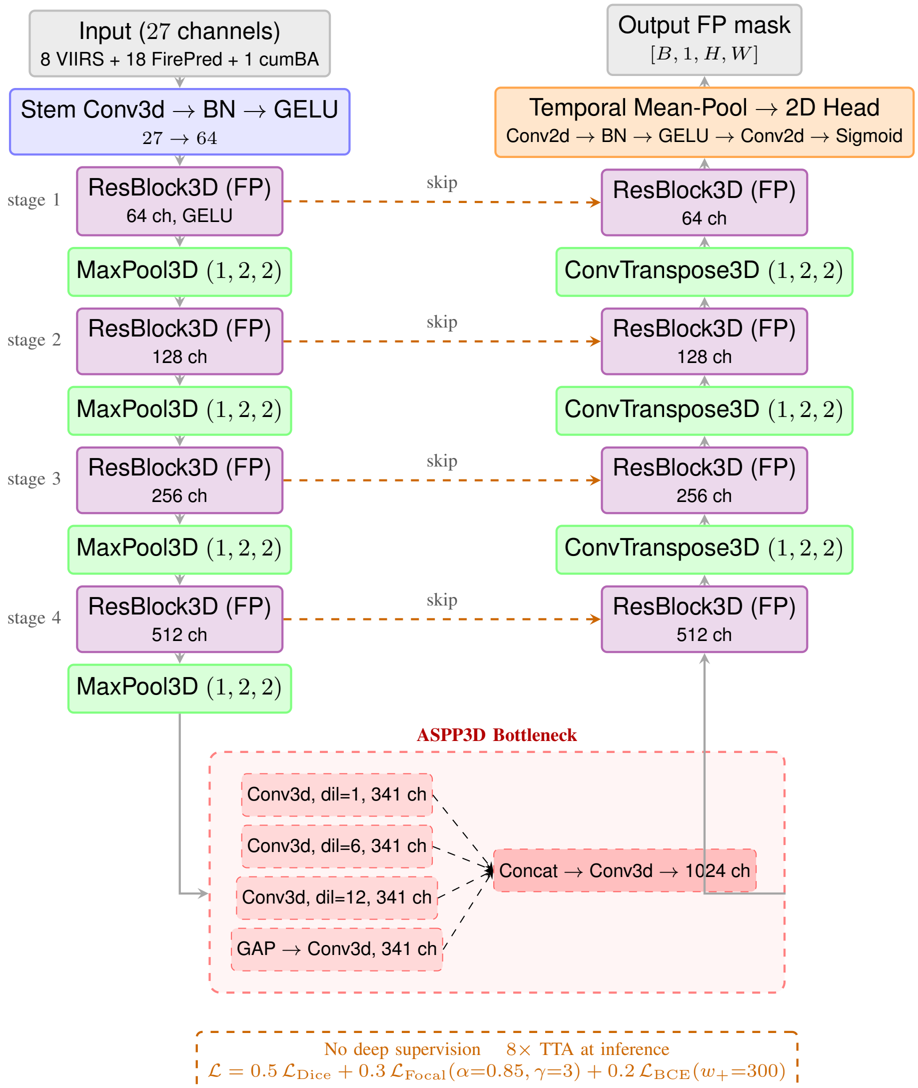


Fig. 3: SpaSE-UNet3D FP variant. The encoder takes a 27-channel input combining VIIRS, FirePred auxiliary bands, and a cumulative burned-area mask. The bottleneck is replaced with an ASPP3D module that aggregates multi-scale context via parallel dilated convolutions and a global average pool. The final head collapses the temporal dimension via a mean-pool and applies a 2D convolutional head to produce a single-day FP mask. No deep supervision is used; $8\times$ test-time augmentation is applied at inference.

Loss.: The FP training objective extends the AF loss with a weighted binary-cross-entropy term:

$$\mathcal{L}_{\text{FP}} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.3\,\mathcal{L}_{\text{Focal}}(\alpha{=}0.85, \gamma{=}3) + 0.2\,\mathcal{L}_{\text{BCE}}(w_+{=}300). \tag{4}$$

The addition of BCE with $\text{pos\_weight} = 300$ is necessary because the FP target is far sparser than AF: typical windows have $< 0.05\%$ positive pixels. The higher Focal $\gamma = 3$ (vs. $\gamma = 2$ in AF) further downweights easy negatives.

Test-time augmentation.: At inference, the FP variant uses $8\times$ test-time augmentation [25]: the input is passed through horizontal flip, vertical flip, both, and three $90°$ rotations, plus the identity, and the resulting logits are averaged before the sigmoid and threshold.

VI. Experiments

A. Implementation Details

All experiments run on a single Kaggle T4 GPU with 16 GB of memory, using PyTorch 2.10 with automatic mixed precision. The training pipeline fits within the 12-hour Kaggle session limit for both tasks.

Shared hyperparameters.: Batch size 8; patch size 256×256 (centre crop of the full scene); optimiser AdamW [26] with learning rate 5×10^{-4} and weight decay 1×10^{-4}; OneCycleLR [27] schedule with $\text{pct\_start} = 0.1$ and cosine annealing; gradient clipping at norm 1.0; random seed 42.

AF.: 80 epochs for TS = 2, 100 epochs for TS = 1. VIIRS channels are normalised with per-band means and standard

TABLE IV: Active Fire detection results on the TS-SatFire test set (15 usable test fires after audit). Paper baselines from [2].

Model	TS	F1	IoU
U-Net (2D)	—	0.731	0.605
Att-U-Net (2D)	—	0.763	0.648
UNETR-2D	—	0.733	0.621
SwinUNETR-2D	—	0.774	0.660
GRU-3	3	0.713	0.601
LSTM-3	3	0.765	0.654
T4Fire	—	0.802	0.700
U-Net-3D	6	0.748	0.628
Att-U-Net-3D	6	0.770	0.654
UNETR-3D	6	0.811	0.706
SwinUNETR-3D	6	0.797	0.688
SwinUNETR-3D	2	0.823	0.727
SpaSE-UNet3D (ours)	1	0.854	0.745
SpaSE-UNet3D (ours)	**2**	**0.855**	**0.746**

deviations computed once on a probe fire and reused across runs.

FP.: 60 epochs. VIIRS channels use the same AF normalisation; FirePred auxiliary bands are normalised with per-band statistics computed at preload time on the training split and cached so that validation and test evaluation use identical statistics. Training data is augmented with horizontal and vertical flips plus $90°$ rotations; test-time augmentation is applied at inference as described in Section V.

Sampling.: For AF we keep only training windows that contain at least 10 fire pixels and include at most 2 times as many negative windows as positive. For FP we use at least 3 burned pixels as the positive threshold and a negative-to-positive ratio of 1.5. These heuristics stabilise training without overfitting to the positive class.

Threshold selection.: For both tasks we sweep the inference threshold over the validation set at 0.05 increments from 0.05 to 0.80, and select the threshold that maximises validation F1. The optimal AF threshold is consistently in the range $[0.20, 0.22]$; the optimal FP threshold is determined per run.

Metrics.: For all tasks we report pixel-level F1 and IoU (Jaccard index) aggregated over all pixels in all test fires (micro-average). Per-fire F1 and IoU are also reported for visualisation.

B. Active Fire Results

Table IV reports the test-set F1 and IoU for SpaSE-UNet3D at TS = 1 and TS = 2, together with the 12 baselines from the dataset paper [2]. SpaSE-UNet3D at TS = 2 achieves F1 = 0.855 and IoU = 0.746, improving the best published baseline (SwinUNETR-3D at TS = 2) by $+3.2\%$ absolute F1 and $+1.9\%$ absolute IoU. The TS = 1 variant is essentially tied with TS = 2 at F1 = 0.854: the addition of a second day of temporal context brings only a $+0.001$ F1 improvement, which supports our design choice of spatial-only convolutions.

Fig. 5 shows the training and validation curves for TS = 2. Validation F1 crosses the paper baseline (dashed red) at approximately epoch 20 and reaches 0.825 by the end of training, for a test F1 of 0.855. Fig. 4 shows the validation threshold sweep, which is flat across $[0.25, 0.69]$ and confirms model robustness to threshold choice. Fig. 6 shows the per-fire F1 distribution: 14 of 15 test fires reach F1 ≥ 0.79, with the single outlier being the low-intensity *Blue Ridge* fire at F1 = 0.746. Fig. 7 shows true-positive (red) / false-positive (green) / false-negative (blue) overlays for all 15 test fires at the optimal threshold.

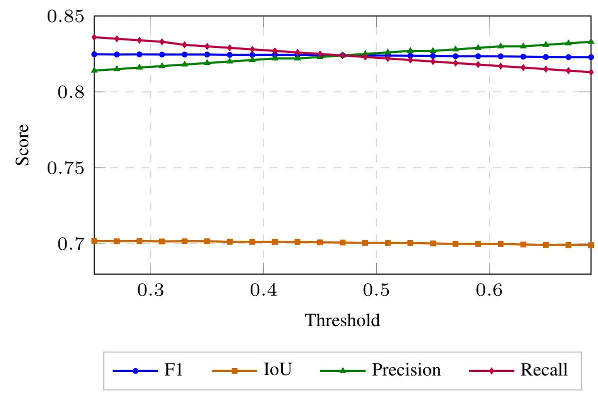


Fig. 4: Active Fire validation threshold sweep at TS = 2. F1 and IoU are nearly flat across $[0.25, 0.69]$; precision and recall trade off symmetrically around 0.50. The model is robust to threshold selection.

Impact of the audit.: To isolate the effect of the label audit, we also trained the same SpaSE-UNet3D architecture on the unaudited training set. The resulting model reaches test F1 = 0.818 (labelled "v1 dirty" in our logs). The audit alone is therefore responsible for a $+3.7\%$ absolute F1 gain, which exceeds the $+3.2\%$ gain of SpaSE-UNet3D over SwinUNETR-3D on audited data. This confirms the argument in Section IV that the label quality of TS-SatFire is a first-class factor in benchmark comparisons.

C. Fire Progression Results

Table V reports the test-set F1 and IoU for SpaSE-UNet3D at TS = 2, together with all FP baselines from the dataset paper. SpaSE-UNet3D reaches F1 = 0.424 and IoU = 0.269, improving the best published baseline (U-Net-3D at TS = 6) by $+4.9\%$ absolute F1. The improvement is particularly notable because SpaSE-UNet3D uses $1/3$ of the temporal context of the best baseline (2 days vs. 6 days). At matched temporal context (TS = 2), SpaSE-UNet3D improves over SwinUNETR-3D (TS = 2) by $+5.8\%$ absolute F1.

Fig. 9 shows the training and validation curves. Fig. 11 shows the validation threshold sweep. Fig. 10 shows the benchmark comparison. Fig. 12 shows the per-fire test F1 distribution. Figs. 13 and 14 show the per-fire prediction maps for all 16 usable US_2021 test fires.

Table VI summarises the resource footprint for both tasks.

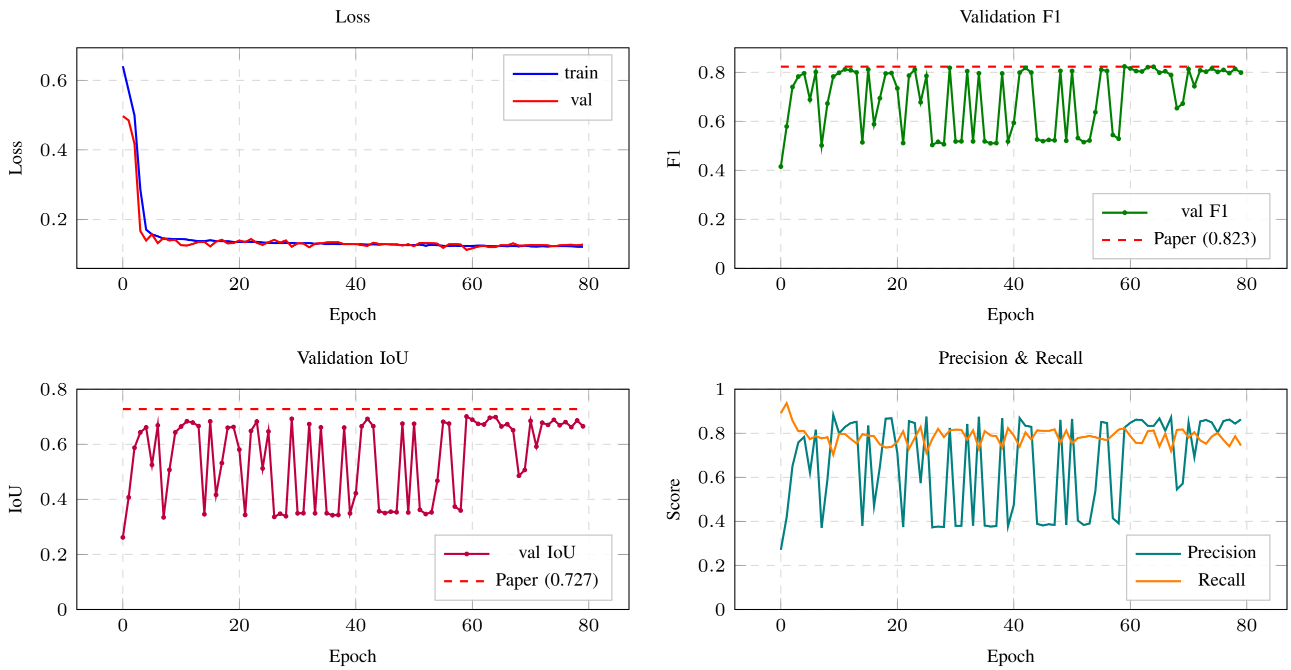


Fig. 5: Active Fire training and validation curves for SpaSE-UNet3D at TS = 2. Loss, validation F1, validation IoU, and precision/recall over 80 epochs. Red dashed lines indicate the published SwinUNETR-3D paper baselines (F1 = 0.823, IoU = 0.727).

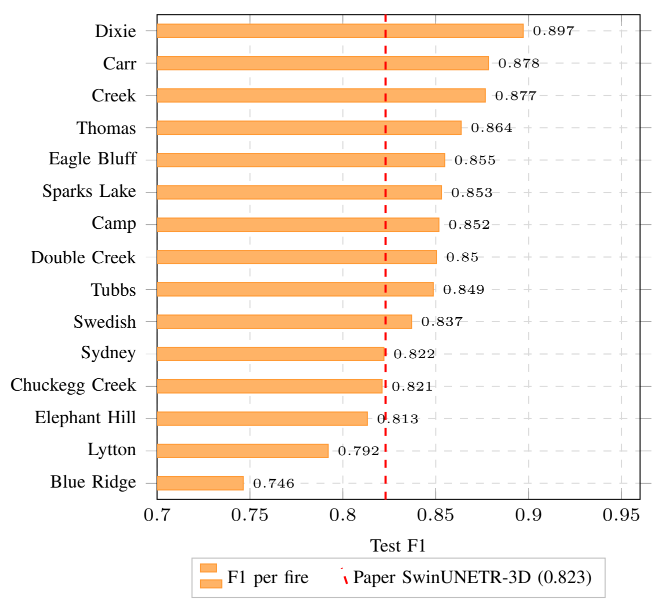


Fig. 6: Per-fire test F1 for SpaSE-UNet3D at TS = 2 on the audited AF test set (15 fires, sorted by F1). Red dashed line: SwinUNETR-3D paper baseline.

TABLE V: Fire Progression prediction results on the TS-SatFire test set (18 usable test fires after AZ/NM exclusion). Paper baselines from [2].

Model	TS	F1	IoU
U-Net-3D	6	0.375	0.338
SwinUNETR-3D	6	0.374	0.331
UNETR-3D	6	0.371	0.336
Att-U-Net-3D	6	0.354	0.312
SwinUNETR-3D	2	0.366	0.321
SpaSE-UNet3D (ours)	**2**	**0.424**	**0.269**

TABLE VI: Training resource footprint on a single Kaggle T4.

Task	Params	Epochs	Time (h)	Peak VRAM
AF (TS = 1)	32.88M	100	4.2	13.2 GB
AF (TS = 2)	32.88M	80	6.5	14.1 GB
FP (TS = 2)	24.28M	60	3.5	13.7 GB

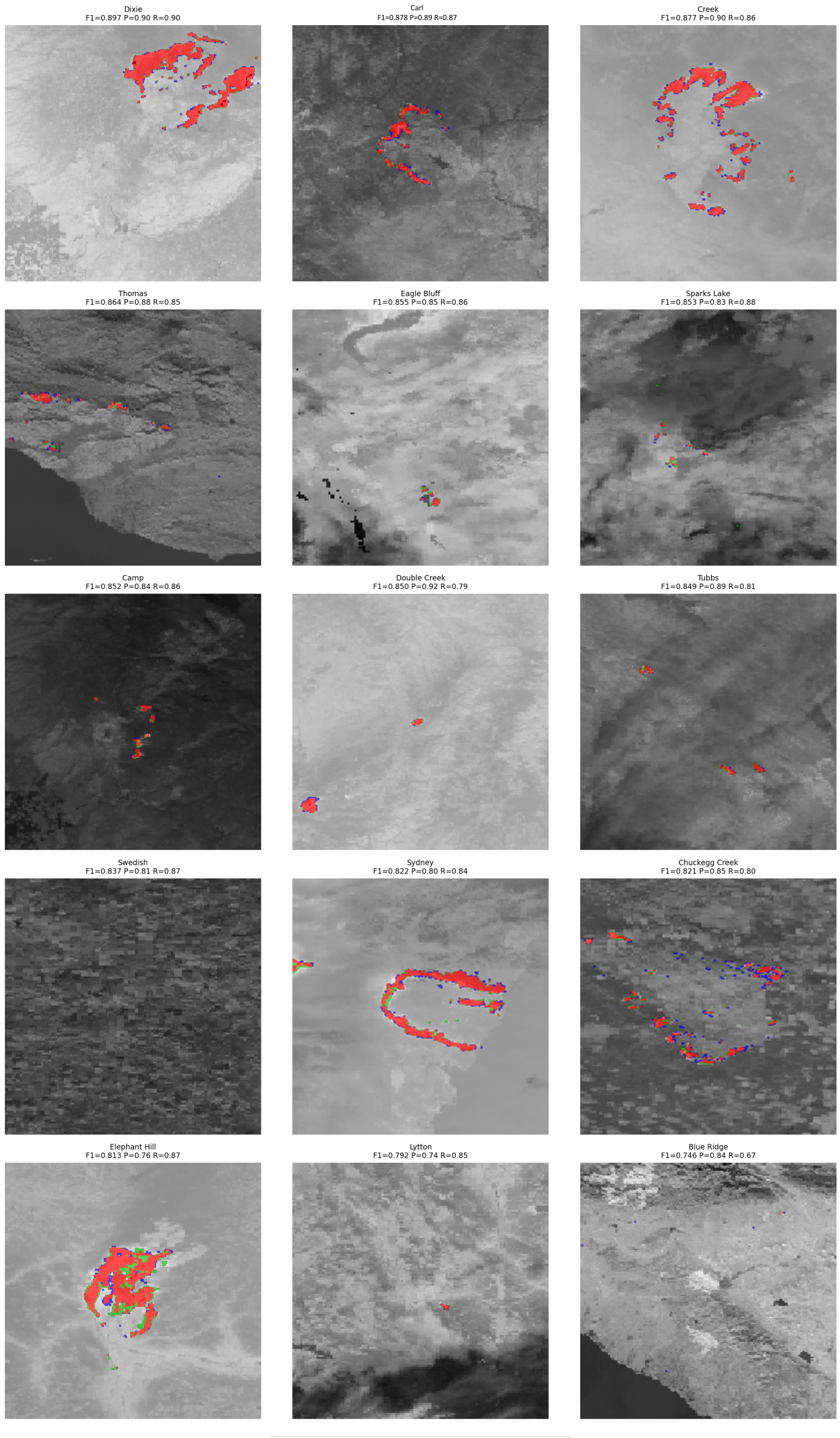


Fig. 7: AF test-set predictions (TS = 2), shown as overlays on the VIIRS day scene. Red: true positive; green: false positive; blue: false negative. Per-fire F1 reported in each panel title.

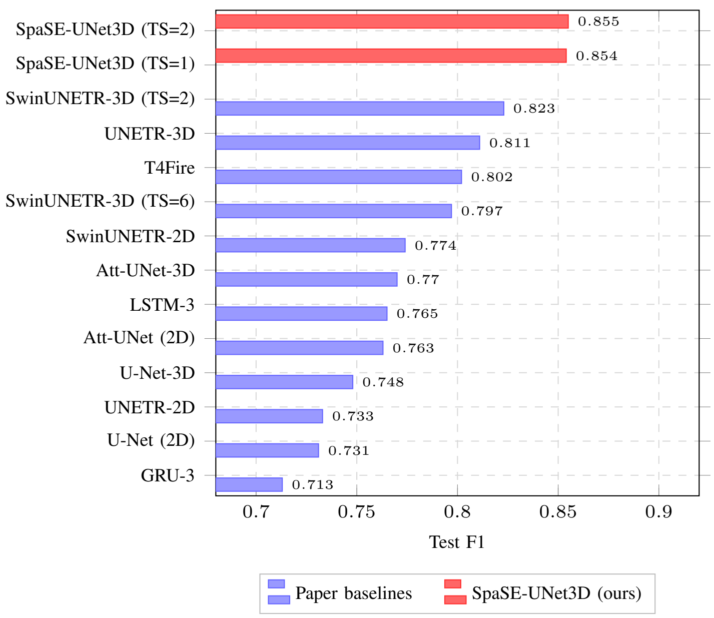


Fig. 8: Active Fire benchmark comparison vs. the 12 paper baselines [2]. Our SpaSE-UNet3D results (TS = 1 and TS = 2) shown in red.

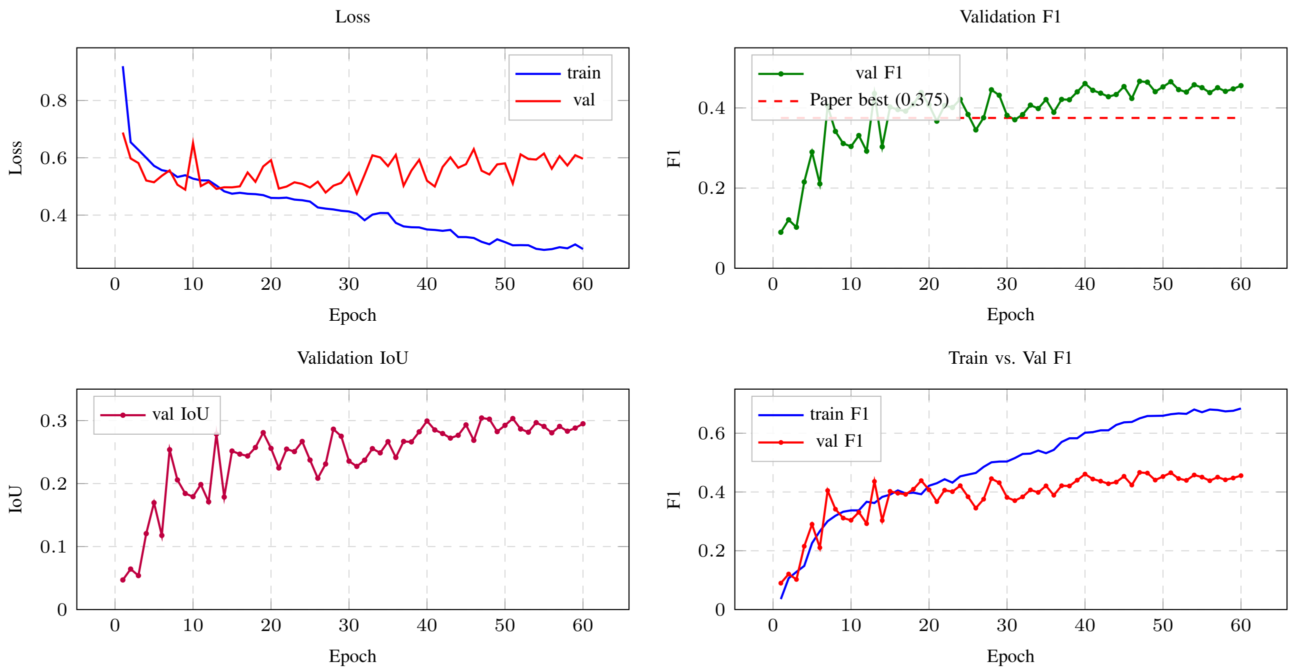


Fig. 9: Fire Progression training and validation curves for SpaSE-UNet3D at TS = 2 over 60 epochs. Red dashed line: best paper baseline (U-Net-3D at TS = 6, F1 = 0.375).

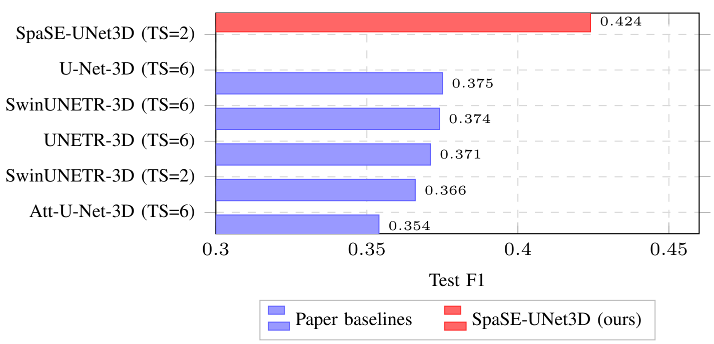


Fig. 10: Fire Progression benchmark comparison. Our SpaSE-UNet3D at TS = 2 shown in red, alongside the five paper baselines [2] at TS = 2 and TS = 6.

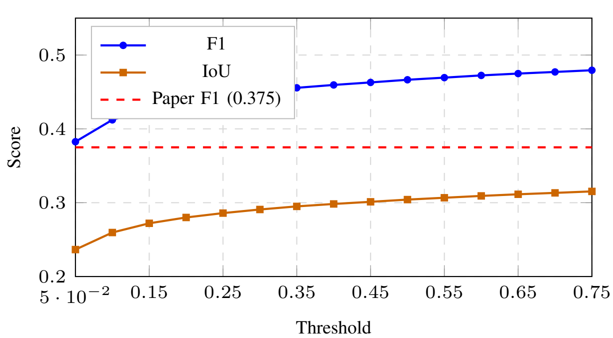


Fig. 11: Fire Progression validation threshold sweep. F1 and IoU as functions of the inference threshold on the validation set. The model is robust over a wide threshold range.

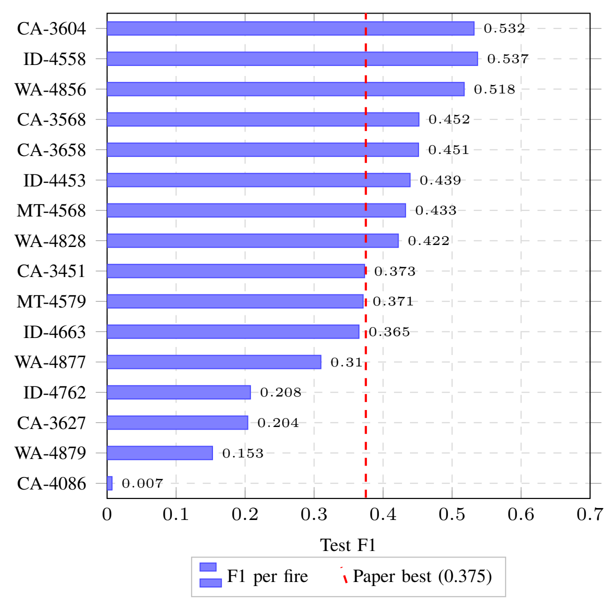
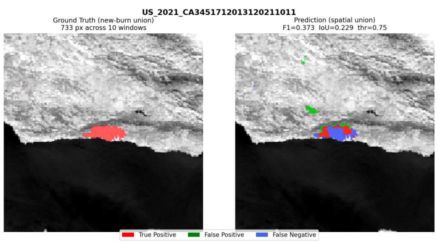
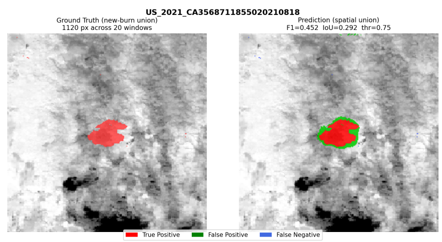


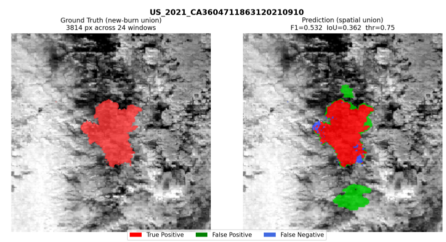
Fig. 12: Per-fire test F1 for SpaSE-UNet3D on the 16 usable US_2021 fires (sorted by F1). Six AZ/NM fires with burn fractions $< 0.001\%$ are excluded as discussed in Section IV. Red dashed line: best paper baseline (0.375).



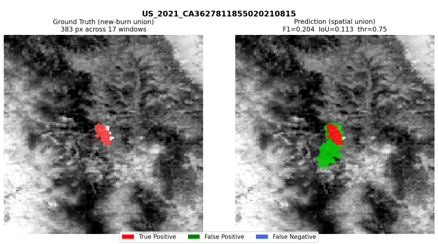
(a) CA 3604 (F1 = 0.532)



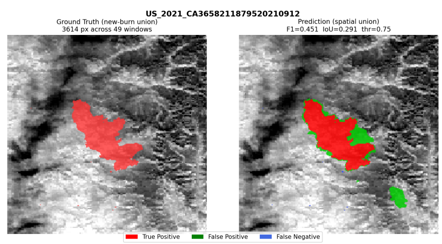
(b) ID 4558 (F1 = 0.537)



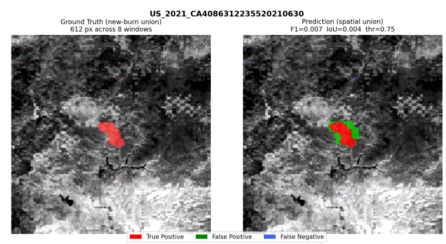
(c) WA 4856 (F1 = 0.518)



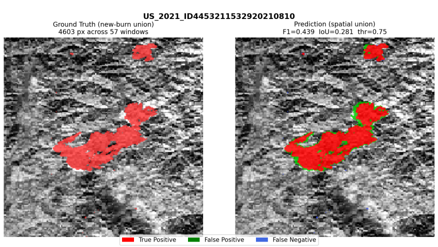
(d) CA 3568 (F1 = 0.452)



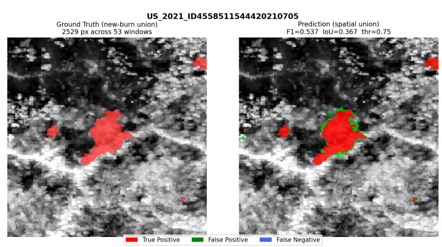
(e) CA 3658 (F1 = 0.451)



(f) ID 4453 (F1 = 0.439)

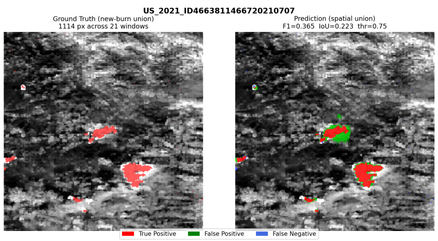


(g) MT 4568 (F1 = 0.433)

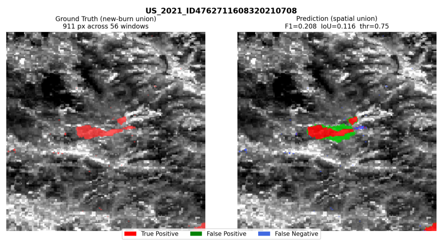


(h) WA 4828 (F1 = 0.422)

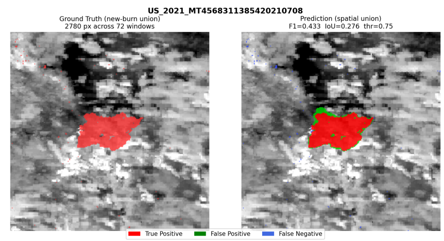
Fig. 13: Per-fire prediction maps (top half: 8 best-performing US_2021 fires). Each panel shows the VIIRS day composite (left) and predicted progression mask overlaid on ground truth (right).



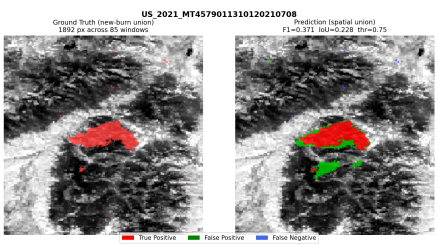
(a) CA 3451 (F1 = 0.373)



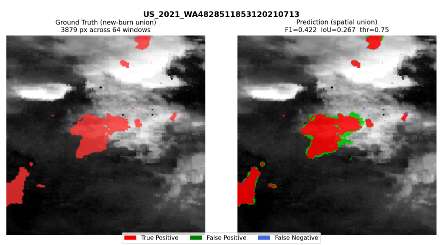
(b) MT 4579 (F1 = 0.371)



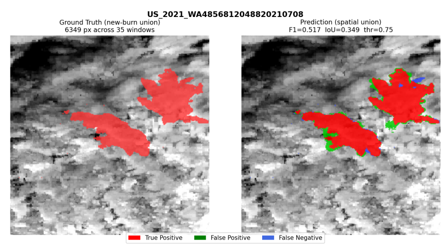
(c) ID 4663 (F1 = 0.365)



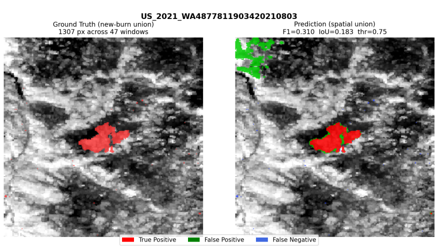
(d) WA 4877 (F1 = 0.310)



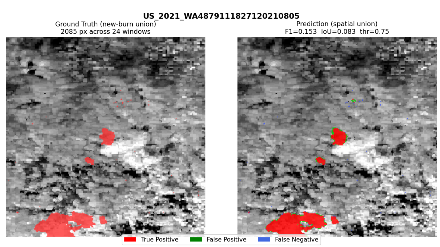
(e) ID 4762 (F1 = 0.208)



(f) CA 3627 (F1 = 0.204)



(g) WA 4879 (F1 = 0.153)



(h) CA 4086 (F1 = 0.007)

Fig. 14: Per-fire prediction maps (bottom half: 8 lower-performing US_2021 fires). Same panel layout as Fig. 13.

VII. Discussion

The results above support three claims that are worth stating plainly.

Data quality is a first-class contribution.: The label audit alone closes $+3.7\%$ absolute F1 on the AF task — a larger delta than our architectural improvement over SwinUNETR-3D on audited data. This is consistent with broader findings that label noise in benchmark test sets routinely destabilises model rankings [18], [19]. Researchers comparing models on TS-SatFire without auditing the labels are therefore measuring data-quality noise as much as they are measuring architectural differences. We encourage the community to treat the exclusion lists released with this paper as a mandatory preprocessing step for the benchmark.

Spatial-only convolutions are the right inductive bias for short temporal stacks.: SpaSE-UNet3D at TS = 1 is essentially tied with TS = 2 (F1 = 0.854 vs. 0.855), and both are well above every 3D baseline at TS $\in \{3, 6\}$. The implicit hypothesis of full 3D convolutions is that temporal neighbours carry complementary information worth mixing early. At TS = 2 this is not true in practice: the temporal information is already captured by the SE-reweighted channel statistics, and dedicating parameters to temporal mixing appears to harm generalisation. This finding is in line with the action-recognition literature where factorised spatiotemporal convolutions outperform full 3D ones [6], [7], and we expect the pattern to generalise to other short-window satellite-remote-sensing tasks.

Task-specific variation is necessary.: The differences between the AF and FP variants — ReLU vs. GELU, Dropout vs. no-Dropout, plain vs. ASPP bottleneck, Dice+Focal vs. Dice+Focal+BCE, 3D vs. 2D-after-temporal-pool head — are not cosmetic. Each was found to matter on its respective task, and the FP objective in particular cannot be trained competitively without the ASPP bottleneck and the BCE term. This suggests that a single fully shared architecture for all three TS-SatFire tasks would give up non-trivial performance on FP.

Limitations.: Our work has three limitations worth noting. First, we do not train a BA model. The BA audit reveals that the most interesting BA modelling question is whether the label encoding (Julian-day codes) can be leveraged for multi-class or regression targets, which we leave to future work. Second, our FP test set excludes the six AZ/NM fires with burn fractions $< 0.001\%$; while we justify this on measurement grounds, some readers may prefer the stricter 24-fire aggregate. Third, our TTA for FP uses only symmetry augmentations; learned or semi-supervised TTA could give a further bump.

VIII. Conclusion

We presented **SpaSE-UNet3D**, a family of spatial-only 3D UNets with Squeeze-and-Excitation channel attention, together with the first systematic label-quality audit of the TS-SatFire benchmark. SpaSE-UNet3D establishes new state-of-the-art results on both the Active Fire detection task ($+3.2\%$ absolute F1 over the best published baseline, at TS = 2) and the Fire Progression task ($+4.9\%$ absolute F1, at $1/3$ the temporal context of the previous best). The label audit is a standalone contribution that exposes substantial label noise in the released benchmark and is required for fair future comparisons. All code, checkpoints, and audit reports are publicly available at https://github.com/mavroul1s/DeepFire-Forecaster.

Acknowledgment

The author thanks the TS-SatFire authors [2] for releasing their dataset and baselines in a form that makes this audit and comparison possible.

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